Mohammad Rahmanitalab

B.Sc. Mechanical Engineering, Sharif University of Technology
Minor in Computer Science, Sharif University of Technology

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Mohammad Rahmanitalab | MRahmaniT

Education

B.Sc. Major in Mechanical Engineering, SUT| Tehran, Iran — *2021 – 2026*
Advisor: Prof. Saeed Behzadipour

- GPA: 3.95/4.00 (18.93/20) — Ranked **12th out of 116** students in the department.

B.Sc. Minor in Computer Science, SUT| Tehran, Iran — *2022 – 2026*

- GPA: 4.00/4.00 (18.98/20)

Diploma in Mathematics and Physics| Tehran, Iran — *2018 – 2021*

- GPA: 4.00/4.00 (19.88/20)

Research Interests

- **Machine Learning:** Deep learning, reinforcement learning
- **Control Systems:** Haptic feedback, teleoperation, master–slave robotic control
- **Medical Robotics:** Surgical robots, exoskeletons, rehabilitation robotics, human–robot
- **Motion Analysis:** Biomechanics, IMU-based sensing, clinical movement classification

Undergraduate Research

Motion Analysis for DMD Rehab | Sharif University of Technology — 2024 – 2026

- Conducting motion analysis using IMU sensors to develop evidence-based therapeutic protocols for children with Duchenne Muscular Dystrophy (DMD), supervised by Prof. Behzadipour.
- Leading the full pipeline: data collection, preprocessing, feature extraction, and movement classification using machine learning models (MATLAB, Python).
- Investigating movement biomarkers to support clinical decision-making in pediatric rehabilitation.

Professional Experience

Back-End Developer | Moj Gostaran Adak, Tehran — Oct – Dec 2025

- Developed a Java-based backend server integrated with the Traccar GPS tracking platform, enabling reliable real-time device communication and data handling.
- Designed and implemented a customized Go backend for the company's internal frontend, supporting a locker-tracking system deployed in the Hamrah Aval project.
- Built optimized RESTful APIs ensuring seamless interoperability between tracking services and client applications.

Internships

Robotics Engineering | Sina Robotics & Medical Innovators Co., Tehran — Jun – Sep 2024

- Worked on surgical robotics with emphasis on kinematics; derived analytical solutions for master–slave robotic systems to improve precision and responsiveness (MATLAB, MSC ADAMS, TwinCAT 3).
- Contributed to the simulation of robotic components, with focus on haptic feedback in teleoperated surgical systems.

Teaching Assistant

Fundamental Automatic Control Design | Sharif University of Technology — Fall 2025

- Assisted in instruction of controller design, PID tuning methods, and stability analysis.
- Managed grading of assignments, quizzes, and exams; held office hours for student guidance.

Automatic Control | Sharif University of Technology — Spring 2025

- Supported lectures on control theory, feedback systems, and Lyapunov stability analysis.
- Assisted with course assessment and provided academic mentoring to students.

Technical Projects

Programming

Advanced Programming (Java) Spring 2025
Developed an online multiplayer game using object-oriented design, multithreading, and socket-based network programming.

Fundamentals of Programming (C) Spring 2022
Implemented a Twitter-like social media application with user management and file-based data persistence.

Robotics and Control

Robotics Fall 2024
Modeled and simulated a multi-DOF robotic manipulator in MATLAB/Simulink, studying forward/inverse kinematics and dynamic behavior.

Fundamental Automatic Control Design Fall 2024
Designed and tuned a PID controller for a ball-and-beam system; analyzed transient response and stability margins in MATLAB/Simulink.

Automatic Control Spring 2024
Designed a PID controller for a rotating governor; performed full system performance and stability analysis.

Dynamics of Machines Spring 2024
Simulated a pushing mechanism in MATLAB/Simulink to analyze forces, motion, and dynamic response.

Skills

Programming: Python, C, Java, Go, TwinCAT 3
Simulation Tools: MATLAB (Simulink), ROS2, Gazebo
Robotics & Control: system modeling, PID control, simulation, sensor data processing
Robot Learning: time-series preprocessing, feature extraction, supervised classification
CAD: SolidWorks, MSC ADAMS
Languages: Persian (Native), English (Excellent)

Awards

High School | Tehran, Iran

- Ranked 187th in the National University Entrance Exam (Konkur). 2021

Undergraduate | Sharif University of Technology

- Ranked 85th in the National Master's Entrance Exam (Konkur). 2025
- Qualified for direct admission to M.Sc. due to academic excellence. 2025 & 2026

Selected Courses

Mechanical Engineering:

- Robotics
- Dynamics of Machinery
- Mechanisms Design
- Automatic Control
- Fund of Automatic Control Design
- Measurement and Control Systems
- Electrical Eng. I–II

Computer Science:

- Advanced Programming
- Dtra Transmission & Networks
- Computer Simulation
- Operating Systems
- Linear Algebra
- Probability
- Discrete Mathematics

Extra-Curricular Activities

CARE Student Society: Served as Media Director, managing media content, production, and team operations.
ICERS 2024: Contributed to event organization and media coordination as a member of the IT and Media Team.
Dept. Website: Developed and maintained the official website of the Mechanical Engineering Department (WordPress).